Chenwei (Denny) Gan

Full-stack software engineer
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Summary

Full-stack software engineer working mainly with TypeScript/React, Node.js, Python/FastAPI, PostgreSQL and AWS. Built and shipped two products end to end: an AWS-hosted web app and an iOS app on the App Store. Master of Information Technology (First Class Honours), University of Auckland, with six years of international business operations experience, which means I ask what a system is for before designing how it works. Post-Study Work Visa, no sponsorship needed.

Technical Skills

Languages
TypeScript, JavaScript, Python, Dart, Swift, SQL, Java
Frontend
React, Vite, Tailwind CSS, React Query, Zustand, Vue.js, Svelte (SvelteKit)
Backend
Node.js, Express, Fastify, FastAPI, SQLModel, Prisma, Zod, REST APIs, WebSocket, OpenAI API
Databases
PostgreSQL, MongoDB (Mongoose), SQLite
Cloud & DevOps
AWS (EC2, RDS, S3, CloudFront, Cognito, Lambda), Terraform, GitHub Actions, Docker, Git
Mobile
Flutter, Riverpod, native Swift (PhotoKit, AVFoundation)
Security
Authentication and authorization (RBAC/ABAC), OAuth, Cognito

Projects

echoType (Web) echotype.ink May 2026 - Present
Solo-built, low-pressure typing web app for practising any texts you love by typing them on a loop, with your own notes shown directly above each phrase, like a gloss written over the line; live since July 2026.

- Anchored notes exactly above their phrase by measuring each character's pixel position in a hidden mirror element, re-measuring only on layout change; a stored snapshot check flags notes that no longer match after the text is edited.
- Defined the AWS infrastructure in Terraform (VPC, EC2, RDS, S3, CloudFront, Cognito, ACM, IAM, SSM, Lambda) and implemented Cognito email/password and Google sign-in with account linking.
- Set up one-click GitHub Actions deploys using OIDC and SSM Run Command, with no inbound SSH: the API runs as a Docker container on EC2 and applies Prisma migrations on startup, and the SPA ships to S3 behind CloudFront.

WakeTok (Mobile) App Store Jul 2026 - Present
Mobile app for rediscovering your own photo and video library by swiping up and down, like the feed in popular short-video apps; designed, built and shipped solo. Free on the App Store since October 2026.

- Flutter UI over a native Swift layer for PhotoKit library access, AVFoundation video playback and Live Photos, connected through Pigeon typed contracts.

- Keeps swiping responsive across full libraries, tested on a real 40,000-item, iCloud-heavy library; saved browsing selections persist locally in Drift/SQLite.
- Product rules (selection, ordering) live in Dart, with layer boundaries enforced by automated architecture tests, keeping the core reusable for a planned Android port.

Professional Experience

Full Stack Developer (Internship) Jul 2025 - Oct 2025

PILOT (Vanta Capital Trading Ltd) | Auckland, New Zealand

Internship at an Auckland fintech startup, building the core infrastructure of a real-time, risk-guardrailed copy-trading MVP, enabling retail users to replicate trades under strict access control and audit requirements.

- Designed and implemented a hybrid RBAC–ABAC authorization model in FastAPI with a three-layer structure spanning user claims, RBAC, and ABAC policy evaluation.
- Built the group-based membership system that governs all interactions between professional traders and retail followers, including following, joining, and trade signal distribution, ensuring every membership action is permission-checked and audit-logged before sensitive trade data reaches any user.
- Independently developed the Trade Order Card system in React and TypeScript with Zustand, creating a single base component with four functional variants and a separate audit layer.
- Implemented a complete RTMP-to-HLS live streaming pipeline (REST + WebSocket + webhook-based session management) so professional traders can broadcast their screens to followers.

GTM (Go-to-Market) Supervisor Nov 2021 - Mar 2023

Tineco | Suzhou, China

Led go-to-market execution across Europe for a home cleaning appliance brand, also leading the global launches of two cordless vacuum series across departments and serving as the business-side owner of several enterprise system rollouts.

- Business-side system owner for the GTM module of the Global Order Management System (GOMS), partnering with IT on requirements gathering, user acceptance testing (UAT), and rollout across European and US subsidiaries.
- Co-defined the functional design of a Java-based internal promotion management system within the GOMS operational ecosystem, replacing an external SaaS tool and supporting multi-channel campaign coordination, conflict detection, and real-time integration with sales and finance workflows.

Event Planning Manager May 2021 - Nov 2021

SUNRA | Wuxi, China

Led nationwide retail campaigns for an e-bike brand, including two promotions across 1,200+ stores, and defined requirements for an online lottery system, built with internal and external IT teams.

Country Retail Manager Jul 2017 - May 2021

Transsion | Dar es Salaam, Tanzania

Led retail operations in Tanzania for the TECNO mobile phone brand, with 300+ staff across 1,000+ stores.

- Turned frontline practice into 34 retail operation standards (SOPs), written from scratch and used nationwide, including supervisor and promoter position manuals and the promotion process.
- Worked with the IT and data teams on the company's retail mobile and web apps, giving feedback on workflows and screen design from the retail side.

Education

University of Auckland 2025

Master of Information Technology

First Class Honours, GPA 8.1/9. Includes an integral industry internship at Pilot.

Jiangxi University of Finance and Economics 2017

Bachelor of Economics, Finance (International Finance)